Adaptations of Marine Mammals

Warm-blooded marine mammals and birds require some special adaptations to permit them to live in ocean waters. One important set of adaptations is concerned primarily with maintaining the animal's temperature. Water has a higher thermal conductivity than air, which means that water is quick to extract heat from a warm body. Humans experience this property when they become chilled after a short time in water of even 80°F(27°C); in air of the same temperature, they would be comfortable indefinitely. Marine mammals, which maintain elevated body temperatures with respect to the surrounding water, must adapt to prevent their body heat from being drained away.

One means of slowing the rate of heat loss is to have a large body. The ratio of the surface area to the volume of any body is lower for a large organism than for a small organism. The larger the body, the smaller the surface area relative to the volume of the body. Since heat production is a function of the volume of a body whereas heat loss is a function of a wetted surface area, this means that large animals, which have a low ratio of surface area to volume, can thermoregulate better than small animals can. Thermoregulation is aided by the smaller surface area that is in contact with the environment through which heat may escape. All swimming marine mammals are large, and it may be that the reason there are no mouse- sized marine mammals is that they would simply die from chilling. There are small marine birds (petrels, auklets), but these animals are completely immersed in the water only for a short time, during intermittent dives. Others, such as gulls, rarely dive, and only a fraction of a bird's body is in contact with the water at any time.

A second adaptation that prevents or reduces heat loss is a thick insulating layer of blubber or fat just beneath the skin. This layer reaches its greatest thickness in whales, where it may be 2 feet thick. In pinnipeds such as the walrus and elephant seal, subcutaneous fat may constitute as much as 33 percent of the weight. Blubber and fat are poor conductors of heat; hence, they protect the animal from losing internal heat. The thicker the blubber or fat layer, the less the heat loss. Marine mammals inhabiting polar waters, therefore, have thicker layers of fat than temperate and tropical species.

A final adaptation concerns the circulatory system. The areas of a marine mammal that offer the greatest surface area to the water (and, therefore, the greatest opportunity for heat loss) and that also lack much of the protective layer of fat are the fins, flippers, and flukes. What adaptations prevent massive heat loss through these extremities? In cetaceans (aquatic mammals), the answer is that the arteries that bring the warm blood out to these extremities are surrounded by a number of smaller veins that bring blood back to the central core of the mammal. Because of this arrangement, the heat of the blood in the arteries can be absorbed by the cooler blood returning in the veins before it is lost to the external water through the thin flesh of the outer extremities. In effect, these animals have a countercurrent system of circulation designed to save heat.

Because most of their adaptations conserve body heat, marine mammals (especially pinnipeds) may, on occasion, become too warm. Hot, still days may mean considerable stress from overheating. On these uncommon occasions, the animals must act to dissipate heat. They do so by waving their flippers in the air while increasing the blood flow to the extremities and restricting the flow back to the core through the veins. The result is heat loss and subsequent cooling. Seals and sea lions may also open their mouths and pant like a dog. If they are resting on land, they may go into the water to cool off.